# The *firmamento* Astronomy ZOO

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## An interactive, data-immersive e-book

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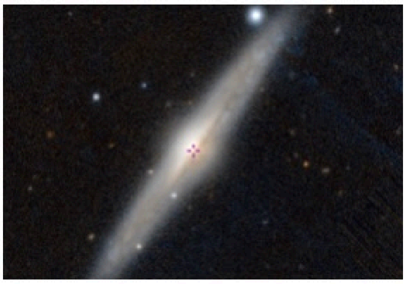
Partly based on AI-generated material
reviewed and curated by experts, and fully immerse in data archives



image credit:

# The Astronomy Zoo and Firmamento



Astronomers often speak of an astronomical zoo to describe the remarkable variety of objects that populate the Universe. Just as a zoological garden brings together many different forms of life, the cosmos contains a vast collection of celestial objects, including stars, planets, nebulae, supernovae remnants, star clusters, galaxies, clusters of galaxies, quasars, black holes, etc. Exploring this astronomical zoo offers insight into the physical processes that shape the Universe and reveals the richness of the cosmic landscape.

The purpose of this e-book is to provide access to high-level astronomical data on a variety of celestial objects through an exploration of modern astronomical archives using the Firmamento platform, accompanied by minimal explanatory text. No prior knowledge of astronomy or of specific scientific concepts is required.

For a deeper involvement in astronomy the reader may refer to the F-book on-line tool or the e-book "An introduction to space and ground based astronomy with Firmamento", which both include a chapter dedicated to public participation to astronomical activities.

## Firmamento data access

Through the Firmamento platform, you can explore spectral and imaging data for a wide range of cosmic sources. Each source type provides an opportunity to investigate the light emitted by these objects, examine their physical characteristics, and discover the information hidden in astronomical observations.

Use the links and icons throughout the following pages to access Firmamento resources, browse object catalogs, and view data for individual sources. Whether you are exploring a specific class of objects or discovering randomly selected examples, these tools allow you to connect directly with real astronomical datasets.

Click the "**Catalog**" icon in the sections below to open a Google Sheet listing the objects in the selected class, along with links to Firmamento spectral and imaging data for each source. Alternatively, click any of the additional icons below to access Firmamento data for randomly selected sources.

# Normal stars

Normal stars are spheres of gas that produce energy through nuclear fusion in their cores, generating an outward pressure that balances the inward pull of gravity due to their mass. Normal stars are grouped by spectral type (O, B, A, F, G, K, M) based mainly on their surface temperature and color, ranging from very hot blue stars to cool red ones. The sequence runs from O-type (extremely hot, massive, and short-lived) through A, F, G, and K types to M-type stars, which are the coolest, smallest, and most common

**Early-type stars (A or B)**

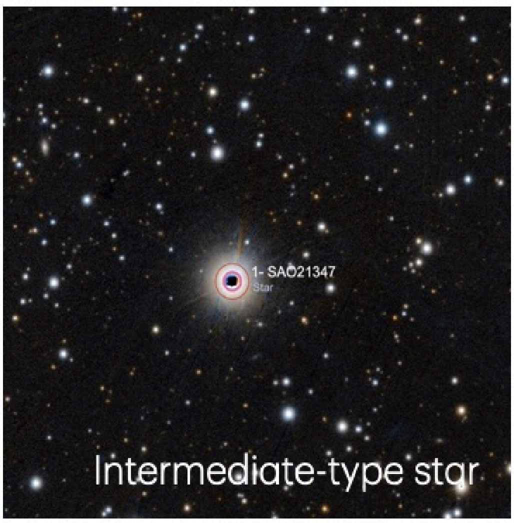


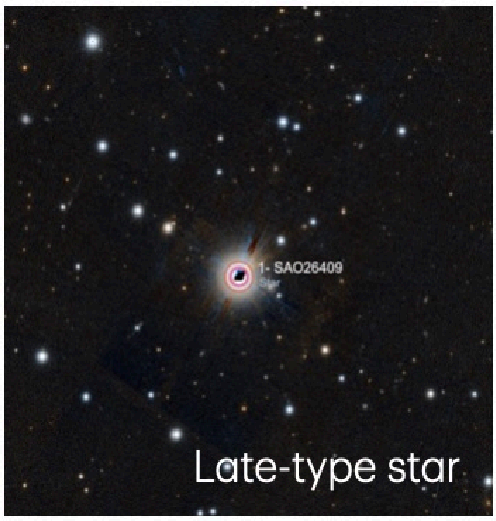
**Intermediate-type stars (F or G)**



**Late-type stars. (K or M)**



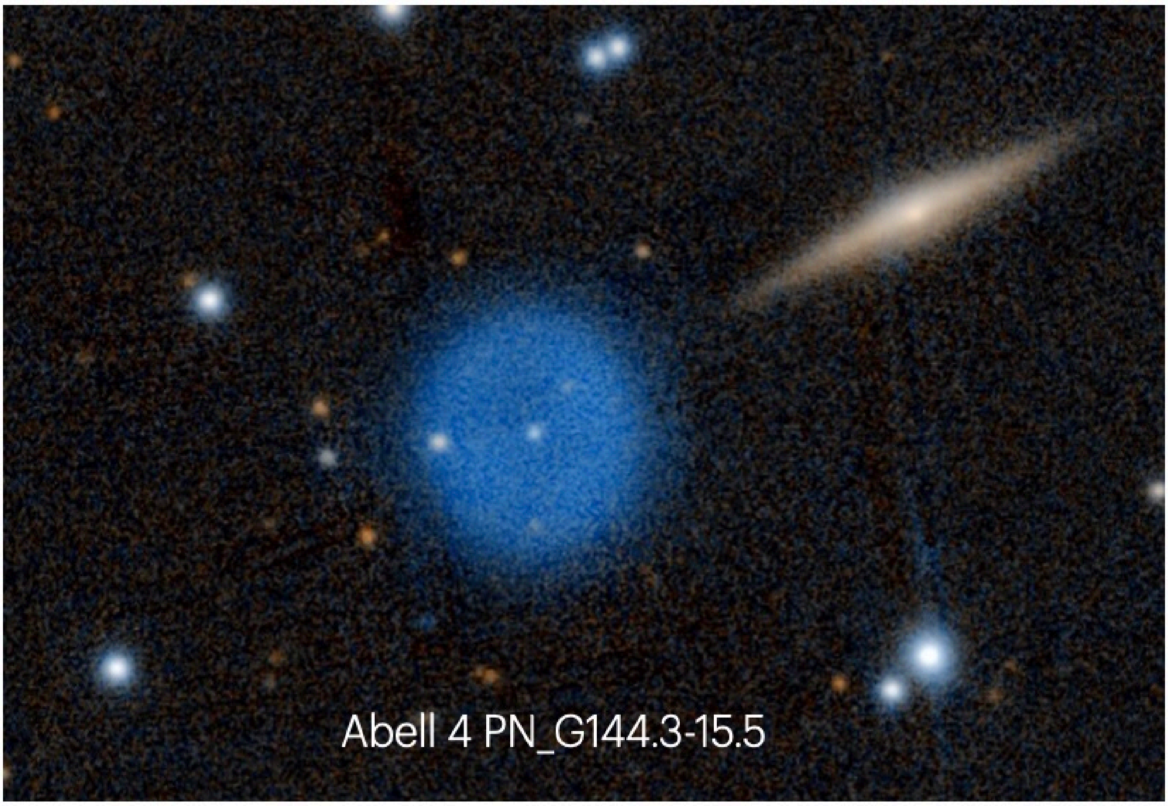



# Planetary nebulae

A planetary nebula is a cloud of glowing gas released by a dying star as it reaches the end of its life, with the gas shining after being heated and excited by the star's hot remnant.





# Pulsars and Pulsar-wind Nebulae

Pulsars are rapidly rotating, highly magnetized neutron stars that emit beams of electromagnetic radiation, observed as periodic pulses as the beams sweep across Earth.

- Radio pulsars: Pulsars that emit beams of radio waves (the majority).

- Gamma-ray pulsars: pulsars whose rotation powers intense gamma-ray emission, often produced in their magnetosphere near the neutron star.

- Pulsar-wind nebulae: glowing clouds of relativistic particles and magnetic fields created when a pulsar's wind interacts with surrounding material.

| Radio Pulsars | Gamma-ray Pulsars | Pulsar Wind Nebulae |
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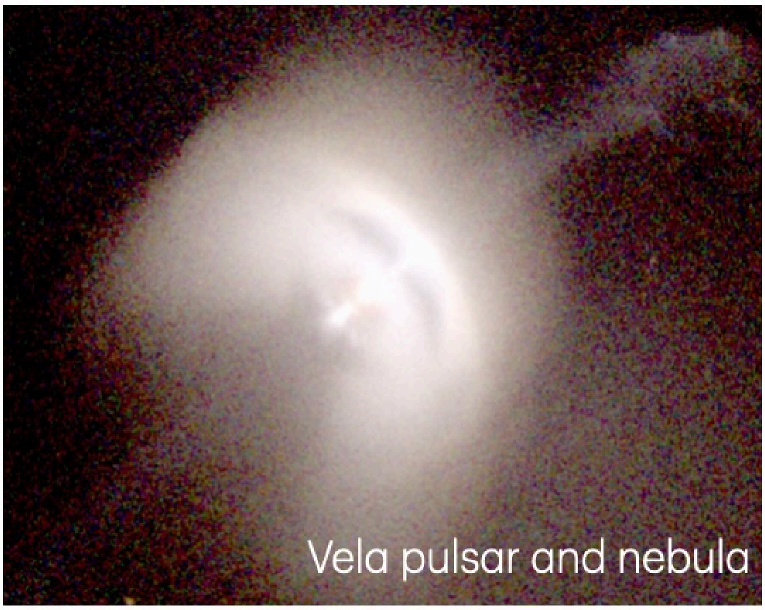


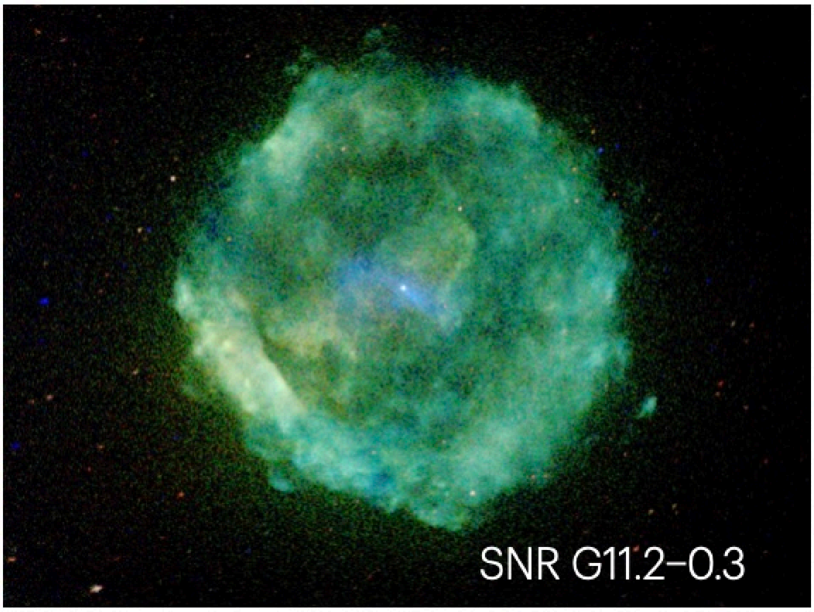

# Normal galaxies

Normal galaxies are large systems including billions of stars, gas, dust, and dark matter held together by gravity, and they are commonly classified into spirals, ellipticals, and irregulars based on their shape and structure. Spiral galaxies have flat rotating disks with spiral arms, elliptical galaxies are more rounded and smooth with little new star formation, and irregular galaxies lack a clear shape.

| Spiral galaxies | Elliptical galaxies | Irregular galaxies |
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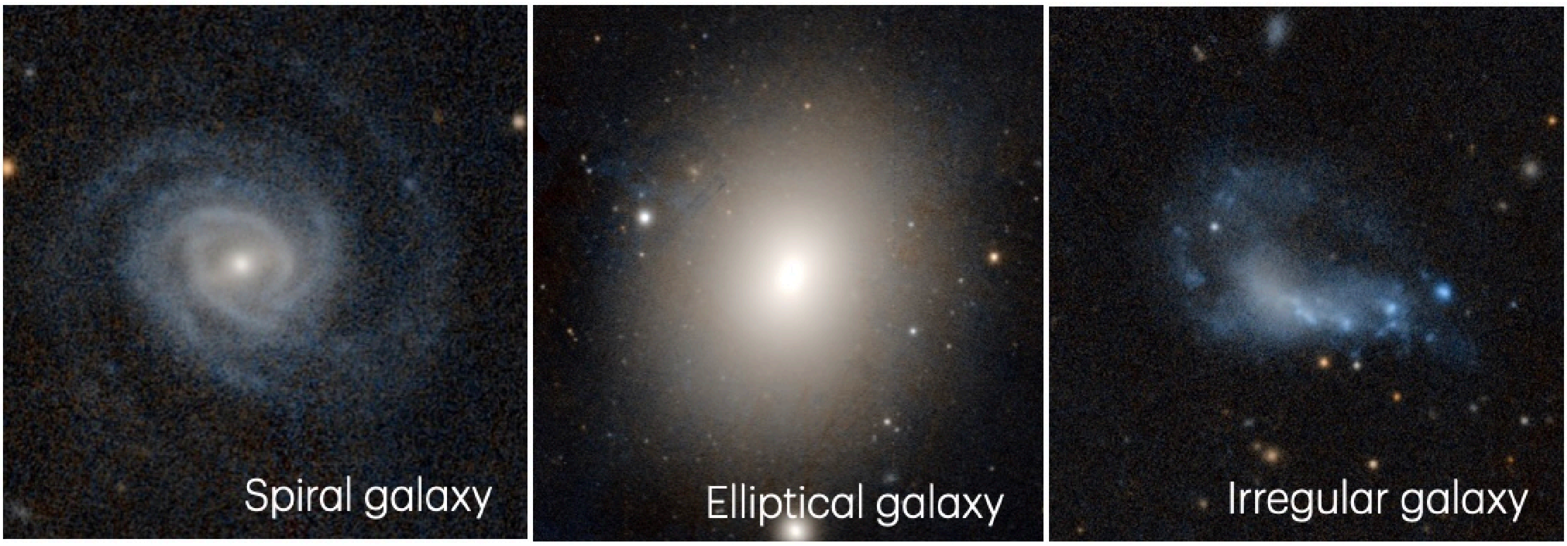

# Clusters of galaxies

Clusters of galaxies are large structures in the universe made up of hundreds to thousands of galaxies bound together by gravity, along with hot gas and dark matter. They are the largest gravitationally bound systems known and often contain more mass in dark matter and hot intracluster gas than in the galaxies themselves.



# Accretion-powered Active Galaxies: Seyfert galaxies and QSOs

Seyfert galaxies are a type of active galaxy with very bright, compact nuclei powered by a supermassive black hole accreting matter, producing strong emission lines. QSOs (quasi-stellar objects) are even more luminous active galactic nuclei that can outshine their entire host galaxies and are visible across vast cosmic distances.

| Seyfert 1 galaxies | Seyfert 2 galaxies | QSO |
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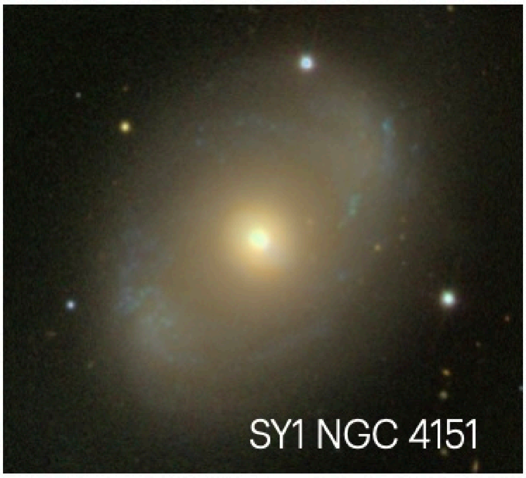
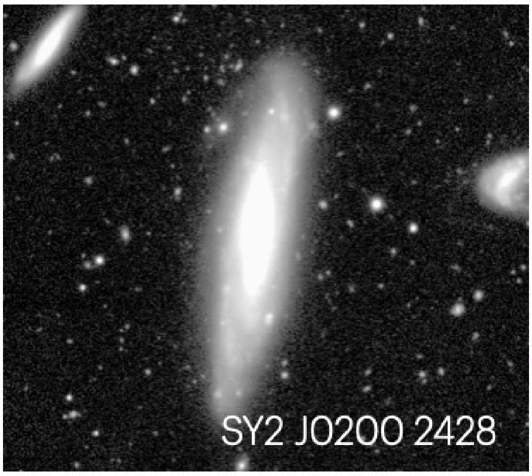
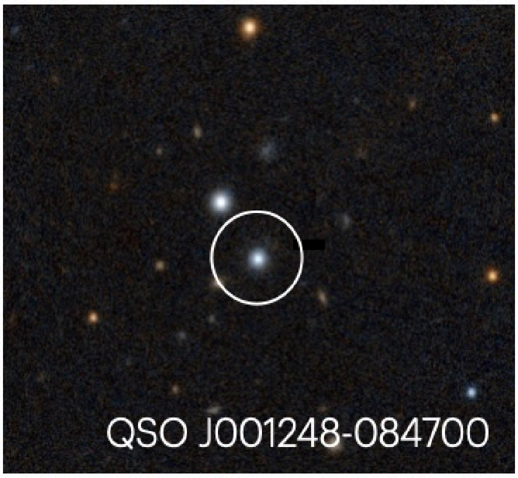




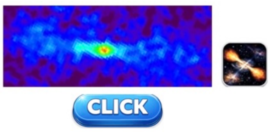


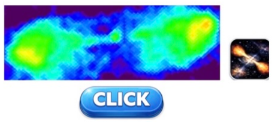
# Jet-powered Active Galaxies: Radio galaxies

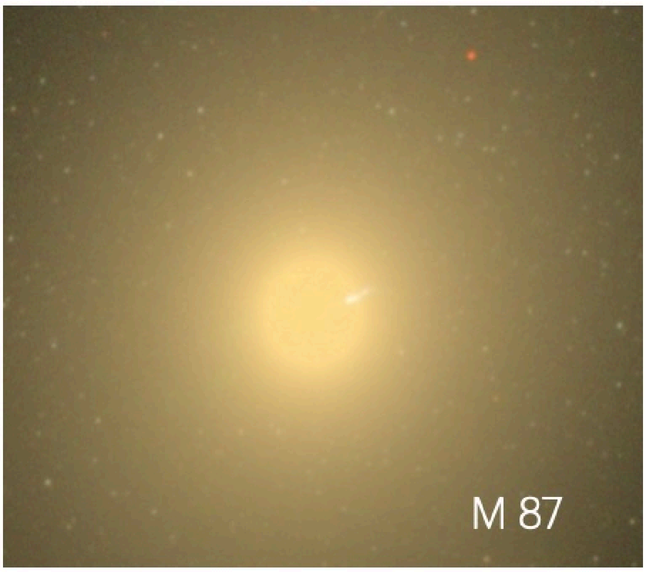
Radio galaxies are active galaxies powered by a supermassive black hole whose relativistic jets produce strong radio emission and large-scale lobes. They are commonly divided into two classes, FR-I and FR-II, based on whether the radio brightness is strongest near the core or at the outer lobes.

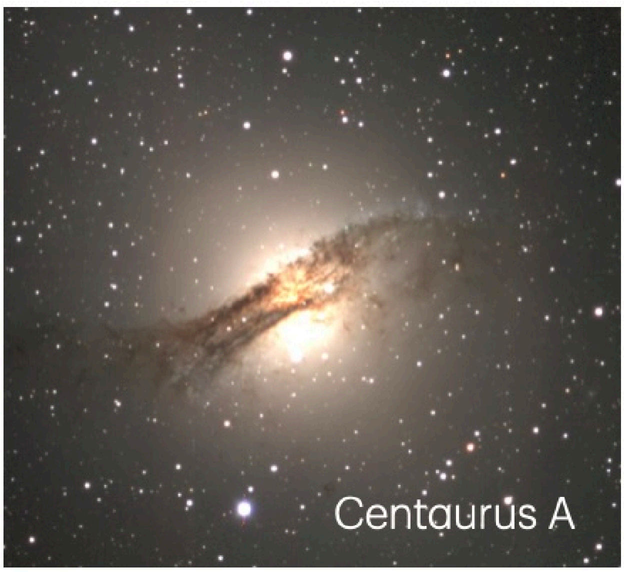
**FR-I Radio Galaxies** **FR-II Radio Galaxies**





# Jet-powered Active Galaxies: Blazars

Blazars are active galactic nuclei with a relativistic jet pointed close to our line of sight, making them highly variable and luminous across the electromagnetic spectrum, and they are classified by their spectral energy distribution into LBL (low-frequency peaked), IBL (intermediate), and HBL (high-frequency peaked) blazars, with some sources detected at very high energies known as TeV blazars.

| LBL Blazars | IBL Blazars | HBL Blazars | TeV Blazars |
|---|---|---|---|